



Brief paper

On functional observability and functional observer design[☆]Mohamed Darouach^{a,*}, Tyrone Fernando^b^a CRAN-CNRS UMR 7039, Université de Lorraine, IUT de Longwy, 186 Rue de Lorraine, 54400 Cosnes-et-Romain, France^b Department of Electrical, Electronic and Computer Engineering, University of Western Australia, 35 Stirling Highway, Crawley, WA 6009, Australia

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ABSTRACT

The objective of this paper is to introduce a new approach to functional observer design. It considers the design of functional observers with pole placement and their application to observer-based control problems. Initially, a new definition of functional observability is provided, followed by algebraic conditions for linear multivariable systems to be functionally observable. These conditions generalize those of observability and the Luenberger observer design in both full and reduced-order cases. The parameters of the functional observer are derived from the constrained Sylvester equation and the functional observability condition. Notably, the observability of the system under consideration is not required for the design of functional observers. The results are applied to the new design of observer-based control without the need to satisfy the observability condition.

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1. Introduction

The problem of observer design, to estimate all the states of a system, has garnered significant attention since the pioneering work of Luenberger (1966, 1971) for deterministic systems and Kalman (1960) for stochastic systems. This issue is of great importance in control and fault diagnosis since typically only the inputs and outputs of the system are available, while many internal states remain inaccessible to direct measurement. On the other hand, for more than five decades, the problem of designing an observer, to estimate a linear combination (function) of the states, has received considerable attention. It can be used in the implementation of control systems, because functional observers can estimate the control signal directly without having to estimate the individual states. Additionally, they can be employed for fault detection and fault-tolerant control (Emami et al., 2015).

Let us consider the following multivariable system,

$$\dot{x}(t) = Ax(t) + Bu(t), \quad (1a)$$

$$y(t) = Cx(t), \quad (1b)$$

$$z(t) = Lx(t), \quad (1c)$$

with the initial state $x(0) = x_0$, $x(t) \in \mathbf{R}^n$ is the state vector, $u(t) \in \mathbf{R}^m$ is the known input, $y(t) \in \mathbf{R}^p$ is the measurement output and $z(t) \in \mathbf{R}^r$ is the functional to be estimated. Matrix $A \in \mathbf{R}^{n \times n}$ and matrix $B \in \mathbf{R}^{n \times m}$. Matrices $C \in \mathbf{R}^{p \times n}$ and $L \in \mathbf{R}^{r \times n}$ are assumed to be of full row rank. The problem to be addressed in this paper is to find an observer for estimating the functional $z(t)$. A functional observer that estimates only one linear function of the state vector, i.e., $r = 1$, was first studied by Luenberger (1966).

Several other works have built upon Luenberger's initial results; specifically, the problem is extended to multifunctional observers or functionals of dimension $r > 1$ in Kimura (1977) and references therein. All of those earlier studies assumed that the considered system is observable. On the other hand, a functional observer of dimension r , which is also referred to as the Darouach observer, having the following form was introduced in Darouach (2000):

$$\dot{\zeta}(t) = N\zeta(t) + Jy(t) + Hu(t), \quad (2a)$$

$$\hat{z}(t) = \zeta(t) + Ey(t), \quad (2b)$$

where the initial state $\zeta(0) = \zeta_0$, $\zeta(t) \in \mathbf{R}^r$ is the state vector of the observer, and matrices $N \in \mathbf{R}^{r \times r}$, $H \in \mathbf{R}^{n \times p}$ and $E \in \mathbf{R}^{r \times p}$ are observer parameters. It has attracted much attention and many results have been developed (see Fernando, Trinh, and Jennings (2010), Rotella and Zambettakis (2016), (Trinh, Tran, & Nahavandi, 2016) and Rotella and Zambettakis (2011)). See also the recent paper Darouach and Fernando (2020) where the authors presented a complete study of these observers. We recall that the necessary and sufficient conditions for existence of an

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observer of dimension r are:

$$\text{rank} \begin{pmatrix} L \\ C \\ CA \\ LA \end{pmatrix} = \text{rank} \begin{pmatrix} L \\ C \\ CA \end{pmatrix}, \quad (3a)$$

$$\text{rank} \begin{pmatrix} \lambda L - LA \\ C \\ CA \end{pmatrix} = \text{rank} \begin{pmatrix} L \\ C \\ CA \end{pmatrix}, \quad (3b)$$

where (3b) needs to be satisfied for all $\lambda \in \mathbb{C}$ for a prescribed rate of convergence of the estimation error. For asymptotic observer design, (3b) needs to be satisfied for all $\lambda \in \mathbb{C}$, $\text{Re}(\lambda) \geq 0$. If these conditions are not met, an observer of dimension $q > r$ that can estimate the functional $z(t)$ may still exist, raising the question of how to design such an observer.

In a recent paper [Darouach and Fernando \(2023\)](#), we presented the definition of functional detectability and the design of an asymptotic functional observer of dimension kr where $k \geq 1$ is an integer scalar. For instance, if $r = 3$, the approach in [Darouach and Fernando \(2023\)](#) considers the design of functional observers of order 3, 6, 9, ..., i.e., only integer multiples of r . Moreover, the approach used in [Darouach and Fernando \(2023\)](#) is based on the static output feedback and LMI approaches. However, in the present paper, we introduce a new definition of functional observability, in the spirit of the definition of full state observability introduced by [Hautus \(1969\)](#). Then, we derive the algebraic conditions for functional observability, they coincide with the observability condition when $r = n$. Contrarily to Ref. [Darouach and Fernando \(2023\)](#), we will consider the design of reduced-order observers of dimension $q = kr + \mu$, where $k, k \geq 1$ and $\mu, 0 < \mu \leq r$, are integer scalars. In this case, if an observer of dimension r does not exist, unlike in [Darouach and Fernando \(2023\)](#), we can consider the design of an observer of dimension $r + 1, r + 2$, etc., by taking $k = 1, \mu = 1, \mu = 2$, etc. The benefit of this approach is that it enables the design of a reduced-order functional observer. Moreover, the proposed pole placement design method is entirely algebraic, allowing precise control over the observer's poles to determine the rate of convergence of the estimation error. This is crucial in observer-based controller design and Ref. [Franklin, Powell, and Emami-Naeini \(1990\)](#) offers practical guidance on selecting observer poles in relation to the controller's poles.

The paper is organized as follows: after the introduction, Section 2 defines functional observability and presents the necessary and sufficient conditions for this type of observability. To establish the connection between functional observability and functional detectability, we first improve the frequency condition given in [Darouach and Fernando \(2023\)](#), which is necessary and sufficient for systems with semi-simple eigenvalues, by considering the more general case where the eigenvalues have different geometric and algebraic multiplicities. Section 3 focuses on the functional observer design, specifically addressing the design of functional observers of order $kr + \mu$ with pole placement. In Section 4, we introduce a new method, based on the results from Sections 2 and 3, for designing an observer-based controller without the need to satisfy the observability condition. To demonstrate the effectiveness of the proposed method, we present two numerical examples. Finally, Section 5 provides the conclusions. In the rest of the paper we shall use the following notation, A^- to denote the generalized inverse of matrix A satisfying only the condition $AA^-A = A$. Also let matrix $A \in \mathbb{R}^{n \times n}$, we denote the set of its n eigenvalues by $\sigma(A)$ and the set of natural numbers including 0 by $\mathbb{N}$. We use α_{ij} to denote the element of the row i and column j of a matrix and introducing the notation J_{ij} for row j of matrix J_i , i.e., $J_i = (J_{i1}^T \ J_{i2}^T \ \cdots \ J_{in}^T)^T$. Also, we will use the following notations: $\Re(\lambda)$ for real part of the complex scalar λ ,

$\text{Ker}(M)$ for the Kernel of matrix M , $\text{Nullity}(M)$ is the nullity of matrix M . Additionally, we will use the following notations:

$$\mathcal{O}_{(M,A,k)} = \begin{pmatrix} M \\ MA \\ \vdots \\ MA^{k-1} \end{pmatrix} \text{ and } \mathcal{O}_{(L,A,k,\mu)} = \begin{pmatrix} \mathcal{O}_{(L,A,k)} \\ L_1 A^k \\ \vdots \\ L_\mu A^k \end{pmatrix}, \text{ where } L_i \text{ is row } i \text{ of matrix } L.$$

2. Functional observability and existence conditions of the functional observer

Now let us consider the linear time invariant system described by Eqs. (1a)–(1c). The solution of (1a) with initial value $x(0) = x_0$ is given by,

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau) d\tau. \quad (4)$$

The corresponding value of the output $y(t, x_0)$ is given by,

$$y(t, x_0) = Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}Bu(\tau) d\tau. \quad (5)$$

We now recall the definition of detectability ([Darouach & Fernando, 2023](#)).

Definition 1. The functional $z(t) = Lx(t)$ is detectable, or the triple (A, C, L) is functional detectable, if the zero input response $y(t) = Cx(t) = 0$ implies that $\lim_{t \rightarrow \infty} z(t) = 0$.

To provide the connection between functional detectability and functional observability, we present new conditions of functional detectability completing those given in [Darouach and Fernando \(2023\)](#), where they concern diagonalizable systems. The following theorem extends those results to a more general case.

Theorem 1. The following conditions are equivalent:

- (1) The functional $z(t) = Lx(t)$ is detectable or the triple (A, C, L) is functional detectable.
- (2) $\text{rank} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \end{pmatrix} = \text{rank} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ \mathcal{O}_{(L,A,n)} \end{pmatrix}, \forall \lambda \in \mathbb{C}, \Re(\lambda) \geq 0.$
- (3) $\text{rank} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \end{pmatrix} = \text{rank} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ L \end{pmatrix}, \forall \lambda \in \mathbb{C}, \Re(\lambda) \geq 0.$

The proof of [Theorem 1](#) is based on the following lemma, which can be directly obtained from Lemma 1 of [Darouach and Fernando \(2023\)](#) by taking the general solution of equation $\dot{x}(t) = Ax(t)$. In fact, if $(A - \lambda I)v = 0$, then $e^{At}v = e^{\lambda t}v$, more generally if $(A - \lambda I)^n v = 0$, then

$$\begin{aligned} x(t) &= e^{At}v = e^{(\lambda I + (A - \lambda I)t)}v, \\ &= e^{\lambda t}(v + t(A - \lambda I)v + \cdots + \frac{t^{n-1}}{(n-1)!}(A - \lambda I)^{n-1}v). \end{aligned} \quad (6)$$

Lemma 1. Assume that the triple (A, C, L) is functional detectable. If $(A - \lambda I)^n v = 0$ (i.e., v is the generalized eigenvector associated to the eigenvalue λ), $v \neq 0$ and $\Re(\lambda) \geq 0$, then $\mathcal{O}_{(L,A,n)}v = 0$.

Now, we can give the proof of [Theorem 1](#).

Proof of Theorem 1. We will prove this theorem as follows, we will first prove that Item (1) of [Theorem 1](#) is equivalent to the

following condition:

$$\begin{aligned} \mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ \mathcal{O}_{(L,A,n)} \end{pmatrix} &= \mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ \mathcal{O}_{(L,A,n)} \end{pmatrix} \\ &= \mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ L \end{pmatrix}, \forall \lambda \in \mathbb{C}, \Re(\lambda) \geq 0, \end{aligned} \quad (7)$$

Then, by using the rank and nullity theorem (Piziak & Odell, 2007), we will deduce that Item (1) is equivalent to Items (2) and (3). Now assume that the triple (A, C, L) is functional detectable and let $v \neq 0$ such that $\begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \end{pmatrix} v = 0$ and $\Re(\lambda) \geq 0$. Then from Lemma 1 we have $\begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(L,A,n)} \end{pmatrix} v = 0$, i.e., for all $\lambda \in \mathbb{C}, \Re(\lambda) \geq 0$,

$$\mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \end{pmatrix} \subseteq \mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ \mathcal{O}_{(L,A,n)} \end{pmatrix}. \quad (8)$$

On the other hand, for all $\lambda \in \mathbb{C}, \Re(\lambda) \geq 0$,

$$\mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ \mathcal{O}_{(L,A,n)} \end{pmatrix} \subseteq \mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ L \end{pmatrix} \subseteq \mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \end{pmatrix}. \quad (9)$$

From Eqs. (8) and (9) we deduce Condition (7), this proves that Item (1) implies Condition (7). To prove that Condition (7) implies Item (1), assume that Condition (7) is satisfied, $y(t) = 0$ and $\lim_{t \rightarrow \infty} z(t) \neq 0, \forall \lambda \in \mathbb{C}, \Re(\lambda) \geq 0$. Then from Eq. (6) we have,

$$\begin{aligned} y(t) &= Cx(t) = e^{\lambda t}(Cv + tC(A - \lambda I)v + \dots \\ &+ \frac{t^{n-1}}{(n-1)!}C(A - \lambda I)^{n-1}v) = 0, \forall \lambda \in \mathbb{C}, \Re(\lambda) \geq 0, \end{aligned}$$

which leads to $v \in \mathcal{Ker} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \end{pmatrix}$, and from Condition (7) we have $v \in \mathcal{Ker}(\mathcal{O}_{(L,A,n)})$. On the other hand by using Eq. (6) we also have,

$$\begin{aligned} \lim_{t \rightarrow \infty} z(t) &= \lim_{t \rightarrow \infty} Lx(t) = \lim_{t \rightarrow \infty} e^{\lambda t}(Lv + tL(A - \lambda I)v + \dots \\ &+ \frac{t^{n-1}}{(n-1)!}L(A - \lambda I)^{n-1}v) = 0, \forall \lambda \in \mathbb{C}, \Re(\lambda) \geq 0, \end{aligned}$$

which contradicts the fact that $\lim_{t \rightarrow \infty} z(t) \neq 0, \forall \lambda \in \mathbb{C}, \Re(\lambda) \geq 0$, this proves that Condition (7) implies Item (1). Thus we have proved that Item (1) and Condition (7) are equivalent. Now, from Condition (7) we have,

$$\text{Nullity} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \end{pmatrix} = \text{Nullity} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ \mathcal{O}_{(L,A,n)} \end{pmatrix} \quad (10)$$

$$= \text{Nullity} \begin{pmatrix} (\lambda I - A)^n \\ \mathcal{O}_{(C,A,n)} \\ L \end{pmatrix}, \forall \lambda \in \mathbb{C}, \Re(\lambda) \geq 0. \quad (11)$$

Using the rank and nullity theorem (Piziak & Odell, 2007), Item (2) is equivalent to Item (3). Thus Theorem 1 is proved. $\square$

Now we can give the following definition.

Definition 2. The functional $z(t) = Lx(t)$, or the triple (A, C, L) is functional observable, if the zero input response $y(t) = Ce^{At}x_0 = 0, \forall t \in [0, T], T > 0$, implies that $z(t) = 0$ on $[0, T]$.

Define the following matrix:

$$\mathbf{P} = \begin{pmatrix} I & 0 & \dots & \dots & 0 \\ -\lambda I & I & 0 & & \vdots \\ (-\lambda)^2 I & -\binom{2}{1}\lambda I & I & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ (-\lambda)^{n-1} I & \binom{n-1}{1}(-\lambda)^{n-2} I & \dots & -\binom{n-1}{n-2}\lambda I & I \end{pmatrix},$$

where $\binom{n}{p} = \frac{n!}{(n-p)!p!}$. The following theorem characterizes the functional observability.

Theorem 2. The following conditions are equivalent:

(1) The functional $z(t) = Lx(t)$ is observable, or the triple (A, C, L) is functional observable.

(2) $\text{rank}(\mathcal{O}_{(C,A,n)}) = \text{rank} \begin{pmatrix} \mathcal{O}_{(C,A,n)} \\ \mathcal{O}_{(L,A,n)} \end{pmatrix}$.

(3) $\text{rank}(\mathcal{O}_{(C,A,n)}) = \text{rank} \begin{pmatrix} \mathcal{O}_{(C,A,n)} \\ LA^i \end{pmatrix}, i \in \mathbb{N}$.

(4) $\text{rank} \begin{pmatrix} C \\ C(A - \lambda I) \\ \vdots \\ C(A - \lambda I)^{n-1} \end{pmatrix} = \text{rank} \begin{pmatrix} C \\ C(A - \lambda I) \\ \vdots \\ C(A - \lambda I)^{n-1} \\ L \end{pmatrix}, \forall \lambda \in \mathbb{C}$.

(5) There exist two proper rational matrices $M_1(\lambda)$ and $M_2(\lambda)$, such that $M_1(\lambda)(\lambda I - A) + M_2(\lambda)C = L$.

Proof. The theorem can be proved as follows: we will prove that Item (1) $\implies$ Item (2) $\implies$ Item (3) $\implies$ Item (2) $\implies$ Item (1). We will also prove that Item (3) $\Leftrightarrow$ Item (4). Item (1) $\Leftrightarrow$ Item (5) can be obtained directly from Hautus (1983). First from (5) for $u(t) = 0$ on $[0, T]$, we obtain $y(t) = Ce^{At}x_0 = \sum_{v=0}^{\infty} \frac{t^v}{v!} CA^v x_0 = 0$, and from Definition 1 we have the functional $z(t)$ observable if $y(t) = Cx(t) = \sum_{v=0}^{\infty} \frac{t^v}{v!} CA^v x_0 = 0 \implies z(t) = Lx(t) = Le^{At}x_0 = \sum_{v=0}^{\infty} \frac{t^v}{v!} LA^v x_0 = 0$ for all t on $[0, T]$, which leads to

$$CA^v x_0 = 0 \implies LA^v x_0 = 0, v \in \mathbb{N}, \quad (12)$$

or using Cayley–Hamilton theorem $\mathcal{O}_{(C,A,n)}x_0 = 0 \implies \mathcal{O}_{(L,A,n)}x_0 = 0$, therefore $\mathcal{Ker}(\mathcal{O}_{(C,A,n)}) \subseteq \mathcal{Ker}(\mathcal{O}_{(L,A,n)})$. So we get Item (1) implies Item (2). If Item (2) is satisfied, then every row of $\mathcal{O}_{(L,A,n)}$ must be in the row space of $\mathcal{O}_{(C,A,n)}$, which proves that Item (2) implies Item (3). Now, if Item (3) is satisfied, then there exist matrices Ω_i of appropriate dimensions such that $L = \sum_{i=1}^{n-1} \Omega_i CA^i$. By induction we have $LA^j = \sum_{i=1}^{n-1} \Gamma_i CA^i$, for $j = 1, n-1$ where Γ_i are matrices of appropriate dimension, which proves that Item (3) implies Item (2). Now, to prove that Item (2) implies Item (1), we assume that Item (2) is satisfied, then there exists a matrix $\mathcal{E}$ of appropriate dimension such that $\mathcal{O}_{(L,A,n)} = \mathcal{E} \mathcal{O}_{(C,A,n)}$. On the other hand, if $y(t) = Ce^{At}x_0 = 0$, we deduce that $\mathcal{O}_{(C,A,n)}x_0 = 0$ and consequently $\mathcal{E} \mathcal{O}_{(C,A,n)}x_0 = \mathcal{O}_{(L,A,n)}x_0 = 0$, which implies that $z(t) = Le^{At}x_0 = \sum_{v=0}^{\infty} \frac{t^v}{v!} LA^v x_0 = 0$ which proves that Item (2) implies Item (1).

On the other hand, Item (3) is equivalent to

$\text{rank}(\mathbf{P}_{\mathcal{O}_{(C,A,n)}}) = \text{rank} \begin{pmatrix} \mathbf{P} & 0 \\ 0 & \mathbf{P} \end{pmatrix} \begin{pmatrix} \mathcal{O}_{(C,A,n)} \\ \mathcal{O}_{(L,A,n)} \end{pmatrix}$, which, since matrix $\mathbf{P}$ is nonsingular, proves that Item (3) is equivalent to Item (4). $\square$

We can give the following corollary from Theorem 2.

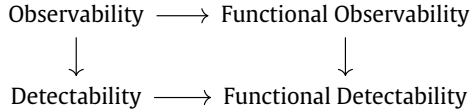
Corollary 1. The functional $z(t) = Lx(t)$ is observable, or the triple (A, C, L) is functional observable, if and only if the observability space from C contains the one from L .

From Theorem 2 we can also give the following remark.

Remark 1.

- (1) When $L = I$, the observer problem is reduced to one of full state estimation, in this case Items (1) and (2) of Theorem 2 become equivalent to the observability of the system (1a) or of the pair (A, C) .
- (2) From Item (3) of Theorem 2, for $i = 0$, the functional observability reduces to the simple condition: $\text{rank}(\mathcal{O}_{(C,A,n)}) = \text{rank}\begin{pmatrix} \mathcal{O}_{(C,A,n)} \\ L \end{pmatrix}$.

The following commutative diagram shows the relations between observability, functional observability, functional detectability and detectability.



In the sequel we make the following assumption:

Assumption 1. The functional $z(t) = Lx(t)$ is observable, i.e., the triple (A, C, L) is functional observable.

Now, for the estimation of the functional $z(t) = Lx(t)$, let us consider the following functional observer,

$$\dot{\hat{\zeta}}(t) = N\hat{\zeta}(t) + Hu(t) + Jy(t), \quad (13a)$$

$$\hat{z}(t) = P\hat{\zeta}(t) + J_1y(t), \quad (13b)$$

with the initial state $\hat{\zeta}(0) = \zeta_0$, $\zeta(t) \in \mathbf{R}^q$ is the state vector of the functional observer and $\hat{z}(t) \in \mathbf{R}^r$ is the estimate of $z(t)$. The matrices N, H, J, P, J_1 are constant of appropriate dimension, which must be determined such that $\lim_{t \rightarrow \infty} [\hat{z}(t) - z(t)] = 0$. For observer (13a) we can give the following definition.

Definition 3. System (13a) is an observer of order q for $z(t) = Lx(t)$ driven by the input $u(t)$ and the output $y(t)$, if $\lim_{t \rightarrow \infty} [\hat{z}(t) - z(t)] = 0$ for every choice of x_0, ζ_0 and the input $u(t)$, and the pair (N, P) is observable.

Remark 2. The observability of the pair (N, P) is necessary to have a minimal realization of system (13a). In Darouach (2000) and Darouach and Fernando (2020) this condition is always satisfied since $P = I$.

Before presenting existence conditions of the functional observer (13a), we give the following definition (Sastry & Bodson, 1989).

Definition 4. Let the system described by the differential equation $\dot{x}(t) = Ax(t)$, then $x = 0$ is called exponentially stable equilibrium point of this system, if there exist scalars $m > 0$ and $\gamma > 0$ such that the solution $x(t)$ satisfies $\|x\| \leq me^{-\gamma t} \|x_0\|$, for all $x_0, t > 0$, where $\|\cdot\|$ is the Euclidian norm and γ is the convergence rate.

Let $\Delta \subset \mathbb{C}$ be any set of appropriate number of eigenvalues (including self-conjugates) in the left half of the complex plane chosen to satisfy the prescribed rate of convergence of the estimation error. Then we can give the following theorem.

Theorem 3. System (13a), with the pair (N, P) observable, is an observer for $z(t)$ with arbitrary rate of convergence of the estimation error, if and only if there exists a matrix $T \in \mathbf{R}^{q \times n}$ such that,

$$NT - TA + JC = 0, \quad (14a)$$

$$PT + J_1C = L, \quad (14b)$$

$$H = TB, \quad (14c)$$

$$\sigma(N) = \Delta \quad (14d)$$

Proof. Let $\epsilon(t) = \zeta(t) - Tx(t)$, then

$$\begin{aligned} \dot{\epsilon}(t) &= \dot{\zeta}(t) - T\dot{x}(t), \\ &= N\epsilon(t) + (NT - TA + JC)x(t) + (H - TB)u(t). \end{aligned} \quad (15)$$

Also from (13b) we have,

$$\hat{z}(t) - z(t) = e(t) = P\epsilon(t) + (PT + J_1C - L)x(t). \quad (16)$$

If conditions (14a)–(14c) are satisfied, then Eqs. (15) and (16) reduce to $\dot{\epsilon}(t) = N\epsilon(t)$ and $e(t) = P\epsilon(t)$ and $e(t)$ converges to 0 with arbitrary rate, if the eigenvalues of matrix N are chosen such that (14d) is satisfied.

Now, for the necessity assume that $e(t)$ converges to 0, i.e., $\lim_{t \rightarrow \infty} e(t) = 0$, then from the assumption of the pair (N, P) is observable, and by using Lemma 2 of Ref. Fortman and Williamson (1972), there exists a matrix T such that $PT + J_1C = L$ and $\lim_{t \rightarrow \infty} \epsilon(t) = 0$, from Definition 4, then the convergence of $e(t)$ to 0 with arbitrary rate of convergence γ , for all $u(t)$ and $x(t)$, implies (14a) and (14d), which proves the Theorem. $\square$

Remark 3. Contrarily to the controllability requirement of the system to establish the necessity of asymptotic observer existence, such a controllability requirement is not needed to establish the necessity to achieve an arbitrary rate of convergence of the estimation error as in Theorem 3. In the case of asymptotic observer design, necessity of Eqs. (14a)–(14c), where the eigenvalues of N are on the left half of the complex plane can be established for systems which are controllable, or more generally for systems with no self-convergent states (i.e., systems with no stable uncontrollable states), see Fernando et al. (2010), (Darouach, 2000), Darouach and Fernando (2020), Darouach and Fernando (2023).

Remark 4. From Theorem 3, we can see that the functional observer design is reduced to solving the Sylvester Eq. (14a) under Constraint (14b) and stability Condition (14d). Condition (14c) can be satisfied once matrix T is determined. It is noted that in the Sylvester Eq. (14a), unlike existing methods for solving the Sylvester equation, J and N are unknown. Additionally, the solution must satisfy Constraints (14b) and (14d), which makes the problem very complex. The following section introduces a new approach to solve this problem. In contrast to our recent work Darouach and Fernando (2023), we will present the design of reduced-order functional observers with pole placement, as opposed to the asymptotic functional observers presented in Darouach and Fernando (2023).

3. Functional observer design

In this section, we assume that the triple (A, C, L) is functional observable. The problem of functional observer design for systems defined by (A, B, C, L) involves determining the matrices T, N, J, H, P , and J_1 . Moreover, if the conditions given by Eq. (3a) are not satisfied, it means that an observer of order $q = r$ does not exist. In this case, we can develop a new method to design an observer of order $q > r$.

Before presenting the new approach to design a reduced-order functional observer, let us consider an observer of order $q = kr + \mu$, where k and μ are integers such that $k \geq 1$ and $0 < \mu \leq r$. The parameters k and μ will be determined in the sequel of the paper. Since the pair (N, P) is observable, we can propose the following observable canonical form for (N, P) :

$$N = \left(\begin{array}{c|c} 0_{(q-r,r)} & I_{q-r} \\ \hline NN_1 & NN_2 \end{array} \right), \quad (17a)$$

$$P = \begin{pmatrix} I_r & | & 0_{(r,q-r)} \end{pmatrix}, \quad (17b)$$

where $NN_1 \in \mathbf{R}^{r \times r}$ and $NN_2 \in \mathbf{R}^{r \times (q-r)}$. Also matrix NN_1 and NN_2 can be written as,

$$NN_1 = \begin{pmatrix} N_{11} & \cdots & N_{1r} \\ \vdots & \ddots & \vdots \\ N_{r1} & \cdots & N_{rr} \end{pmatrix} \text{ and } NN_2 = \begin{pmatrix} N_{1(r+1)} & \cdots & N_{1q} \\ \vdots & \ddots & \vdots \\ N_{r(r+1)} & \cdots & N_{rq} \end{pmatrix}.$$

Now we can consider two cases: (i) $0 < \mu < r$ and $r > 1$, (ii) $\mu = r$ or $r = 1$.

3.1. Case where $0 < \mu < r$ and $r > 1$

Let matrices T, J, J_1 and L be as follows,

$$T = ((T_1)^T \ (T_2)^T \ \cdots \ (T_r)^T \ (T_{r+1})^T \ \cdots \ (T_q)^T)^T, J = ((J_2)^T \ (J_3)^T \ \cdots \ (J_{r+1})^T \ (J_{r+2})^T \ \cdots \ (J_{q+1})^T)^T, J_1 = ((J_{11})^T \ (J_{12})^T \ \cdots \ (J_{1r})^T)^T,$$

and $L = ((L_1)^T \ (L_2)^T \ \cdots \ (L_r)^T)^T$,

where $T \in \mathbf{R}^{q \times n}$, $J \in \mathbf{R}^{q \times p}$, $J_1 \in \mathbf{R}^{r \times p}$ and $L \in \mathbf{R}^{r \times n}$.

From Eq. (14b) we deduce the following matrices,

$$\begin{aligned} T_1 &= L_1 - J_{11}C, \\ T_2 &= L_2 - J_{12}C, \\ &\vdots \\ T_r &= L_r - J_{1r}C, \end{aligned} \quad (18)$$

and from Eq. (14a) we deduce,

$$\begin{aligned} T_{r+1} &= T_1A - J_2C = L_1A - J_{11}CA - J_2C, \\ T_{r+2} &= T_2A - J_3C = L_2A - J_{12}CA - J_3C, \\ &\vdots \\ T_q &= T_{q-r}A - J_{q-r+1}C, \end{aligned} \quad (19)$$

and

$$\begin{pmatrix} N_{11} & \cdots & N_{1q} \\ N_{21} & \cdots & N_{2q} \\ \vdots & \ddots & \vdots \\ N_{r1} & \cdots & N_{rq} \end{pmatrix} T - \begin{pmatrix} T_{q-r+1} \\ \vdots \\ T_q \end{pmatrix} A + \begin{pmatrix} J_{q-r+2} \\ \vdots \\ J_{q+1} \end{pmatrix} C = 0. \quad (20)$$

From Eq. (18), for $j = 1, \dots, r$, we have,

$$T_j = L_j - J_{1j}C, \quad (21)$$

and from Eq. (19), we deduce the following recurrence:

$$T_{ir+j} = T_{(i-1)r+j}A - J_{(i-1)r+j+1}C, \quad (22)$$

for $k > 1$ and $i = 1, \dots, k-1$, the term j varies from 1 to r , and for $k \geq 1$ and $i = k$, the term j varies from $j = 1$ to μ . From Eqs. (21) and (22) we obtain the following relations for matrices T_j ,

$$\begin{aligned} T_{ir+j} &= L_jA^i - J_{1j}CA^i - J_{j+1}CA^{i-1} \\ &\quad - J_{j+1+r}CA^{i-2} - \cdots - J_{(i-1)r+j+1}C, \end{aligned} \quad (23)$$

where for $k > 1$ and $i = 1, \dots, k-1$, the term j varies from 1 to r , and for $k \geq 1$ and $i = k$, the term j varies from $j = 1$ to μ .

We can see from Eqs. (21) and (23) that matrix T can be expressed in terms of matrices J_1 and J . Now, for $l = 1, \dots, r$, let

$$N_l = (N_{l1} \ N_{l2} \ \cdots \ N_{lq}), \quad (24)$$

then Eq. (20) can be written as

$$N_lT - T_{q-r+l}A + J_{q-r+l+1}C = 0, \quad (25)$$

where $q = kr + \mu$. Since matrix T can be expressed in terms of J_1 and J according to (21) and (23), we can rewrite (25) as follows:

$$N_lO_{(L,A,k,\mu)} + \Phi_lO_{(C,A,k+1)} = L_{\mu+l}A^k, \quad l = 1, \dots, r - \mu \quad (26)$$

and for $l = r - \mu + \psi$, $\psi = 1, \dots, \mu$,

$$N_lO_{(L,A,k,\mu)} + \Phi_lO_{(C,A,k+2)} = L_{\psi}A^{k+1}, \quad (27)$$

where

$\Phi_l = (\Phi_{l1} \ \Phi_{l2} \ \cdots \ \Phi_{l(k+1)})$, for $l = 1, \dots, r - \mu$ and $\Phi_l = (\Phi_{l1} \ \Phi_{l2} \ \cdots \ \Phi_{l(k+2)})$, for $l = r - \mu + \psi$, with $\psi = 1, 2, \dots, \mu$. Moreover, see Appendix A for the definitions of matrices Φ_l . Matrices N_l in (26) and (27) can be obtained as follows, let $\mathbf{L}$ be the set of the desired poles of the observer and define the following matrices $\bar{N}$ and $\bar{B}$:

$$\bar{N} = \left(\begin{array}{c|c} 0_{(q-r,r)} & I_{q-r} \\ \hline 0 & 0 \end{array} \right), \quad (28a)$$

$$\bar{B} = \left(\begin{array}{c} 0_{(q-r,r)} \\ \vdots \\ I_{1r}^T \end{array} \right)^T. \quad (28b)$$

Since the pair $(\bar{N}, \bar{B})$ is controllable, we can always find a gain matrix $\bar{K}$ such that the eigenvalues of $\bar{N} - \bar{B}\bar{K}$ can be placed in the set $\mathbf{L}$. Then we can determine N by assigning matrix $N = \bar{N} - \bar{B}\bar{K}$, and considering the structure of N according to 17a, we can determine N_l considering (24). Now since matrices N_l are known, Eqs. (26) and (27) can be rewritten as follows:

$$\Phi_lO_{(C,A,k+1)} = L_{\mu+l}A^k - N_lO_{(L,A,k,\mu)}, \quad l = 1, \dots, r - \mu. \quad (29)$$

and for $l = r - \mu + \psi$, $\psi = 1, \dots, \mu$,

$$\Phi_lO_{(C,A,k+2)} = L_{\psi}A^{k+1} - N_lO_{(L,A,k,\mu)}. \quad (30)$$

Conditions for existence of solutions, Φ_l , to Eqs. (29) and (30) are:

$$\begin{aligned} \text{rank} \begin{pmatrix} L_{\mu+l}A^k - N_lO_{(L,A,k,\mu)} \\ O_{(C,A,k+1)} \end{pmatrix} &= \text{rank } O_{(C,A,k+1)}, \\ l &= 1, \dots, r - \mu \end{aligned} \quad (31)$$

and

$$\begin{aligned} \text{rank} \begin{pmatrix} L_{\psi}A^{k+1} - N_lO_{(L,A,k,\mu)} \\ O_{(C,A,k+2)} \end{pmatrix} &= \text{rank } O_{(C,A,k+2)}, \\ l &= r - \mu + \psi, \psi = 1, \dots, \mu. \end{aligned} \quad (32)$$

Under Conditions (31) and (32), the general solutions to Eqs. (29) and (30) are given by:

$$\begin{aligned} \Phi_l &= (L_{\mu+l}A^k - N_lO_{(L,A,k,\mu)})O_{(C,A,k+1)}^{-1} \\ &\quad - Y_{l1}(I - O_{(C,A,k+1)}O_{(C,A,k+1)}^{-1}), \text{ for } l = 1, \dots, r - \mu \end{aligned} \quad (33)$$

and for $l = r - \mu + \psi$, with $\psi = 1, \dots, \mu$

$$\begin{aligned} \Phi_l &= (L_{\psi}A^{k+1} - N_lO_{(L,A,k,\mu)})O_{(C,A,k+2)}^{-1} \\ &\quad - Y_{l2}(I - O_{(C,A,k+2)}O_{(C,A,k+2)}^{-1}), \end{aligned} \quad (34)$$

where Y_{l1} and Y_{l2} are arbitrary matrices of appropriate dimensions, which can be set equal zero. Once Φ_l is determined, observer matrices J_1 and J can be determined using equations in Appendix B, and matrix T can be determined by (18) and (19), and observer matrix H is determined by (14c). Observer matrices $P = (I_r \ 0)$ and $N = \bar{N} - \bar{B}\bar{K}$.

3.2. Case where $\mu = r$ or $r = 1$

For this case we have $q = (k+1)r$, then matrix N of 17a can be written in the following block form

$$N = \left(\begin{array}{c|c} 0_{(kr,r)} & I_{kr} \\ \hline \mathcal{N} & \end{array} \right), \quad (35)$$

where matrix $\mathcal{N} = (\mathcal{N}_1 \ \mathcal{N}_2 \ \cdots \ \mathcal{N}_{k+1}) \in \mathbf{R}^{r \times (k+1)r}$. Let matrices T and J be partitioned according to the partition of matrix $\mathcal{N}$, then we have $T = (\mathcal{T}_1^T \ \mathcal{T}_2^T \ \cdots \ \mathcal{T}_{k+1}^T)^T$ and $J = (\mathcal{J}_2^T \ \cdots \ \mathcal{J}_{k+2}^T)^T$, where $T \in \mathbf{R}^{(k+1)r \times n}$ and $J \in \mathbf{R}^{(k+1)r \times p}$.

Now from (14b) we deduce $\mathcal{T}_1 = L - J_1 C$, and from (14a) we deduce,

$$\mathcal{T}_i = \mathcal{T}_{i-1} A - J_i C, i = 2, \dots, k+1. \quad (36)$$

Using the value of $\mathcal{T}_1$ we deduce,

$$\mathcal{T}_i = LA^{i-1} - J_1 CA^{i-1} - \mathcal{J}_2 CA^{i-2} - \dots - \mathcal{J}_i C, i = 2, \dots, k+1, \quad (37)$$

and

$$\sum_{i=1}^{k+1} \mathcal{N}_i \mathcal{T}_i - \mathcal{T}_{k+1} A + \mathcal{J}_{k+2} C = 0. \quad (38)$$

By inserting (37) into (38) we obtain the following,

$$\mathcal{N} \mathcal{O}_{(L,A,k+1)} + \Psi \mathcal{O}_{(C,A,k+2)} = LA^{k+1}, \quad (39)$$

where $\Psi = (\Psi_1 \quad \Psi_2 \quad \dots \quad \Psi_{k+2})$ with

$$\Psi_i = \mathcal{J}_{(k+2)-i+1} - \sum_{j=i}^{k+1} \mathcal{N}_j \mathcal{J}_{j-i+1}, i = 2, \dots, k+1, \quad (40)$$

and

$$\Psi_{k+2} = J_1. \quad (41)$$

Now, as in Section 3.1 we first find matrix $\mathcal{N}$ by using the pole placement of matrix $N = \tilde{N} - \tilde{B}\tilde{K}$, where $\tilde{N} = \begin{pmatrix} 0_{(kr,r)} & I_{kr} \\ 0_{(r,(k+1)r)} \end{pmatrix}$.

Then (39) can be written as

$$\Psi \mathcal{O}_{(C,A,k+2)} = LA^{k+1} - \mathcal{N} \mathcal{O}_{(L,A,k+1)}. \quad (42)$$

From this equation, matrix Ψ exists if and only if,

$$\text{rank} \begin{pmatrix} LA^{k+1} - \mathcal{N} \mathcal{O}_{(L,A,k+1)} \\ \mathcal{O}_{(C,A,k+2)} \end{pmatrix} = \text{rank} \mathcal{O}_{(C,A,k+2)}. \quad (43)$$

Under Condition (43) the general solution to (42) is,

$$\Psi = (LA^{k+1} - \mathcal{N} \mathcal{O}_{(L,A,k+1)}) \mathcal{O}_{(C,A,k+2)}^{-1} - Z(I - \mathcal{O}_{(C,A,k+2)} \mathcal{O}_{(C,A,k+2)}^{-1}), \quad (44)$$

where Z is an arbitrary matrix which can be taken equal 0. Once matrix Ψ is determined, we can determine the parameter J_1 from (42). Then, by using a backward recursion starting from $i = k+1$ to $i = 2$ in (40) to determine the parameter the components $\mathcal{J}_i$ of matrix J .

Remark 5. Given that the triple (A, C, L) is functional observable, then there exist scalars k and μ such that Conditions (31) and (32) are satisfied, and also there exists a scalar k such that (43) is satisfied.

Now we can summarize the presented approach for the functional observer design. First, check if the triple (A, C, L) is functional observable. If so, then proceed with the design, otherwise stop. Given that the triple (A, C, L) is functional observable, first for $q = r$, check if conditions given by Equations (3a) and (3b) are satisfied, if so then an observer of dimension r exists and the approach given in Darouach (2000) can be used to design this observer.

For the design of functional observers of dimension $q = kr + \mu > r$ we can give the following procedure:

- Step 1: Let $k = 1$.
- Step 2: Let $\mu = 1$.
- Step 3: The observer is of dimension $q = kr + \mu$. Let $\mathbf{L}$ be the set of the desired poles, then use any pole placement method to compute matrix N by using Eqs. (28a) and (28b). If $r = 1$, go to Step 5, otherwise, go to Step 4.

- Step 4: If Conditions (31) and (32) are satisfied, then go to Step 6, otherwise set $\mu = \mu + 1$, if $\mu = r$, go to Step 5, otherwise go to Step 3.
- Step 5: If Condition (43) is satisfied, an observer of dimension $q = (k+1)r$ exists, then go to Step 7, otherwise set $k = k+1$ and go to Step 2.
- Step 6: Deduce the values of matrices Φ_i according to (33) and (34), then from Appendix B compute matrices J_1 and J and deduce matrix T from (21)–(22).
- Step 7: Deduce the values of matrices Ψ_i according to (44), then compute matrices J_1 from (42), and find the elements of matrix J by using a backward recursion starting from $i = k+1$ to $i = 2$ in (40), then deduce matrix T from (37).

Now let us consider the following example:

Example 1.

Consider the linear system (1a) described by the following

$$\text{matrices: } A = \begin{pmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & -1 & 1 & 0 \\ 0 & 0 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & -4 & -2 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 2 & 1 \\ 1 & 0 \\ 4 & 1 \end{pmatrix},$$

$$C = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \text{ and } L = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

The system is unobservable, however, it is functional observable and Conditions of Eqs. (3a) and (3b) for the existence a functional observer of dimension $q = r = 2$ are not satisfied. On the other hand, from the results of Darouach and Fernando (2023), one can see that an asymptotic observer of dimension $q = 2r = 4$ exists. Now, from the presented results we find for $k = 1$ and $\mu = 1$, an observer of dimension $r+1 = 3$ exists. For a set of desired eigenvalues $\mathbf{L} = \{-2, -4, -5\}$, we find the following matrices: $N = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -5 & 0 \\ -8 & 0 & -6 \end{pmatrix}$, $J = \begin{pmatrix} 0 & -10 \\ 4 & -4 \\ 0.3686 & 39.6314 \end{pmatrix}$, $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$,

$$J_1 = \begin{pmatrix} 0.3686 & 2.6314 \\ -2 & 2 \end{pmatrix}, \text{ and } H = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 0.7373 & 1.3686 \end{pmatrix}.$$

For the simulation we take as inputs: $u_1 = 3\text{Heaviside}$ and $u_2 = 2\text{Heaviside}$. For this simulation we take as initial state for the system $x_0 = (0 \ 0 \ 0.5 \ 0.3 \ 0.05 \ 0.2)^T$ and as initial state for the observer $\zeta_0 = (0.5 \ 0.8 \ 0.2)^T$. The obtained estimates are given by Fig. 1. This Figure shows the good convergence of these estimates $\hat{z}_i(t)$, $i = 1, 2$ to the components $z_i(t)$, $i = 1, 2$ of $z(t)$.

4. Observer-based controller design

In this section we shall show how the presented results can be used in the design of an observer-based controller even when the considered system is unobservable. We shall show how the functional observer can be used in the closed-loop instead of the full or reduced-order observer and that the observability of the pair (A, C) is not required. First, assume that the pair (A, B) is controllable and let L be a stabilizing gain for system (1a)–(1b). Also assume that the triple (A, C, L) is functional observable. Then, from the presented results we can design a functional observer of the form (13a)–(13b), with poles assigned in a desired position, estimating directly $z(t) = Lx(t)$. Now, let us apply the following control law $u = -\hat{z}(t) + u_r$ to system (1a)–(1b), where u_r is the reference input. Then, we have the following theorem.

Theorem 4. The characteristic polynomial of the composite system formed by system (1a) and the observer (13a) is the product of the characteristic polynomials of matrices $A - BL$ and N .

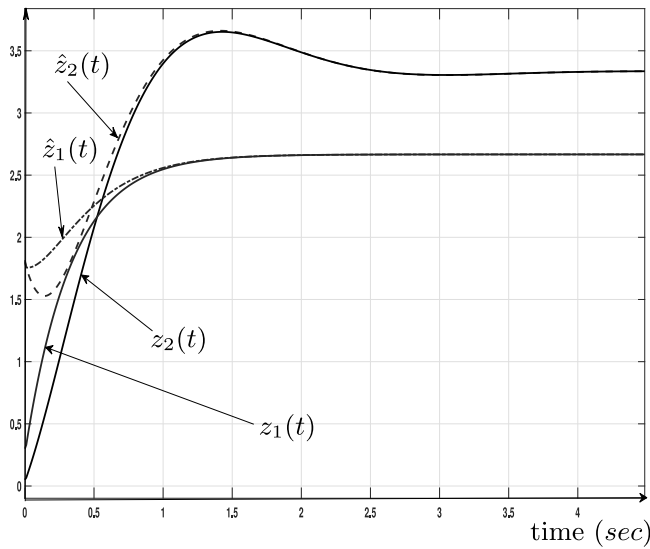


Fig. 1. Functionals $z_1(t)$ and $z_2(t)$ and their estimates.

Proof. By inserting the value $u = -\hat{z}(t) + u_r$ in system (1a) we obtain

$$\dot{x}(t) = Ax(t) - B\hat{z}(t) + Bu_r. \quad (45)$$

By using Eq. (13b), we have the following equation

$$\begin{aligned} \dot{x}(t) &= Ax(t) - BP\zeta(t) - BJ_1y(t) + Bu_r, \\ &= (A - BJ_1C)x(t) - BP\zeta(t) + Bu_r. \end{aligned} \quad (46)$$

On the other hand from (13a) we obtain

$$\dot{\zeta}(t) = (N - HP)\zeta(t) + (J - HJ_1)Cx(t) + Hu_r. \quad (47)$$

Now, from (46) and (47) we obtain the following interconnected system obtained from the open-loop system (45) and the observer-based controller:

$$\begin{pmatrix} \dot{x}(t) \\ \dot{\zeta}(t) \end{pmatrix} = \begin{pmatrix} A - BJ_1C & -BP \\ (J - HJ_1)C & N - HP \end{pmatrix} \begin{pmatrix} x(t) \\ \zeta(t) \end{pmatrix} + \begin{pmatrix} B \\ H \end{pmatrix} u_r. \quad (48)$$

Recall the following errors defined in the proof of Theorem 3, $\epsilon(t) = \zeta(t) - Tx(t)$ and $e(t) = \hat{z}(t) - z(t) = P\epsilon(t)$ and by applying the coordinate transformation $\begin{pmatrix} x(t) \\ \epsilon(t) \end{pmatrix} = \begin{pmatrix} I & 0 \\ -T & I \end{pmatrix} \begin{pmatrix} x(t) \\ \zeta(t) \end{pmatrix}$ to Eq. (48), we obtain

$$\begin{pmatrix} \dot{x}(t) \\ \dot{\epsilon}(t) \end{pmatrix} = \begin{pmatrix} A - BL & -BP \\ 0 & N \end{pmatrix} \begin{pmatrix} x(t) \\ \epsilon(t) \end{pmatrix} + \begin{pmatrix} B \\ 0 \end{pmatrix} u_r. \quad (49)$$

From this equation we can see that the characteristic polynomial of the closed-loop system is the product of the characteristic polynomials of matrices $A - BL$ and N , which proves the theorem. $\square$

From Eq. (31) we can see that the closed-loop system is asymptotically stable if and only if matrices N and $A - BL$ are Hurwitz stable. In this case the dimension of the closed-loop system is $q+n \leq 2n-p$. We can also remark that the design of the functional observer is dependent on the functional observability of the triple (A, C, L) , and we have the following corollary.

Corollary 2. Let L be a matrix such that matrix $A - BL$ is Hurwitz stable with arbitrary pole placement. Then the sufficient condition to have the separation principle independently of the gain matrix L is that the pair (A, C) must be observable.

Proof. It is easy to see that if the pair (A, C) is observable, then the triple (A, C, L) is functional observable and we can design a functional observer, with a pole placement, independently of the matrix $A - BL$. $\square$

Remark 6. Corollary 1 implies that if the system is observable, it is also functional observable, allowing us to design an observer of dimension $q < n - p$, instead of estimating the entire state with a full-order observer.

Now we will present a numerical example to illustrate the results presented in this section.

Example 2.

Here we consider example 1 of Section 3 for observer-based controller design. One can see that this system is unobservable and controllable. Then from the existing results on the full and reduced-order observers design (Trinh & Fernando, 2011), it is not possible to design an observer-based controller. Since the pair (A, B) is controllable, then there exists a gain matrix L that place the poles of $A - BL$ in an arbitrary region. It is also easy to see that the matrix $L = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$

places the poles of the state feedback system at $\sigma(A - BL) = (-1 - 2 - 3 - 4.2 - 0.9021 + 0.7850i - 0.9021 - 0.7850i)$. On the other hand, the triple (A, C, L) is functional observable and dimension of matrix L , $r = 2$ does not exist. Following the approach presented in Section 3, we find an observer of dimension 3. Let the set of the desired poles of the observer be $\mathbf{L} = (-5 \ -4 \ -1)$. Then obtain the following matrices

$$\begin{aligned} T &= \begin{pmatrix} 0 & -2.9351 & -6.0649 & 0 & 1 & 0 \\ 0 & -2 & 0 & 1 & 0 & 1 \\ 0 & 17.6108 & 40.3892 & 2.9351 & -8 & 1 \end{pmatrix}, \\ N &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ -20 & 0 & -9 \end{pmatrix}, \quad J_1 = \begin{pmatrix} 2.9351 & 6.0649 \\ 2 & 0 \end{pmatrix}, \\ J &= \begin{pmatrix} -8.8054 & -31.1946 \\ 4 & -2 \\ 46.9621 & 179.0379 \\ 2 & 0 \end{pmatrix} \text{ and } H = TB = \begin{pmatrix} 1 & 0 \\ 6 & 2 \\ 1.8703 & 3.9351 \end{pmatrix}. \end{aligned}$$

This example shows that even if the considered system is unobservable, with the concept of functional observability and the functional observer design, we can design a reduced-order observer-based controller.

5. Conclusion

In this paper a new definition of functional observability and an original method are proposed for the design of reduced-order functional observers. The existence of the functional observer of a general form was presented and completes the presented results in Darouach and Fernando (2023) by the pole placement of the observer. The obtained results were applied to observer based control without satisfying the observability condition.

Appendix A

By inserting Eqs. (21) and (23) into Eq. (25) we obtain Eqs. (26)–(27) with the following values of Φ_l :

(1) For $\mu < r$, $l = 1, \dots, r - \mu$:

$$\Phi_{l1} = J_{(k-1)r+\mu+l+1} - N_l \begin{pmatrix} J_{11} \\ J_{12} \\ \vdots \\ J_{(k-1)r+\mu+1} \end{pmatrix}, \quad \Phi_{l2} =$$

$$J_{(k-2)r+\mu+l+1} = (N_{l(r+1)} \quad \cdots \quad N_{1q}) \begin{pmatrix} J_{11} \\ J_{12} \\ \vdots \\ J_{(k-2)r+\mu+1} \end{pmatrix},$$

$$\Phi_{l3} = J_{(k-3)r+\mu+l+1} - (N_{l(2r+1)} \quad \cdots \quad N_{1q}) \begin{pmatrix} J_{11} \\ J_{12} \\ \vdots \\ J_{k-3r+\mu+1} \end{pmatrix},$$

$$\cdots \Phi_{l(k+1)} = J_{1(\mu+l)} - N_{l(kr+1)}J_{11} - N_{l(kr+2)}J_{12} -$$

$$\cdots - N_{1q}J_{1\mu} \text{ and } \Phi_{lk} = J_{\mu+l+1} - N_{l((k-1)r+1)}J_{11} -$$

$$\cdots - N_{l(kr)}J_{1r} - N_{l(kr+1)}J_{12} - \cdots - N_{1q}J_{\mu+1}.$$

(2) For $\mu < r$, $l = r - \mu + \psi$ with $\psi = 1, 2, \dots, \mu$:

$$\Phi_{l1} = J_{kr+\psi+1} - (N_{l1} \quad N_{l2} \quad \cdots \quad N_{rq}) \begin{pmatrix} J_{11} \\ J_{12} \\ \vdots \\ J_{(k-1)r+\mu+1} \end{pmatrix},$$

$$\Phi_{l2} = J_{(k-1)r+\psi+1} - (N_{lr(r+1)} \quad \cdots \quad N_{rq}) \begin{pmatrix} J_{11} \\ J_{12} \\ \vdots \\ J_{(k-2)r+\mu+1} \end{pmatrix},$$

$$\Phi_{l3} = J_{(k-2)r+\psi+1} - (N_{l(2r+1)} \quad \cdots \quad N_{1q}) \begin{pmatrix} J_{11} \\ J_{12} \\ \vdots \\ J_{(k-3)r+\mu+1} \end{pmatrix},$$

$$\cdots, \Phi_{l(k+1)} = J_{\psi+1} - N_{1q}J_{1\mu} \text{ and } \Phi_{l(k+2)} = J_{1\psi}$$

Appendix B

From [Appendix A](#) we can deduce sequentially the values of the matrices J_1 and J as follows:

$$J_{11} = \Phi_{(r-\mu+1)(k+2)}$$

$$\vdots$$

$$J_{1\mu} = \Phi_{r(k+2)}$$

$$J_{1(\mu+1)} = \Phi_{1(k+1)} + N_{1(kr+1)}J_{11} + N_{1(kr+2)}J_{12}$$

$$+ \cdots + N_{1q}J_{1\mu}$$

$$J_{1(\mu+2)} = \Phi_{2(k+1)} + N_{2(kr+1)}J_{11} + N_{2(kr+2)}J_{12}$$

$$+ \cdots + N_{2q}J_{1\mu}$$

$$\vdots$$

$$J_{1r} = \Phi_{(r-\mu)(k+1)} + N_{(r-\mu)(kr+1)}J_{11}$$

$$+ \cdots + N_{(r-\mu)q}J_{1\mu}$$

$$J_2 = \Phi_{(r-\mu+1)(k+1)} + N_{r-\mu+1}J_{1\mu}$$

$$J_3 = \Phi_{(r-\mu+2)(k+1)} + N_{(r-\mu+2)q}J_{1\mu}$$

$$\vdots$$

$$J_{\mu+1} = \Phi_{r(k+1)} + N_{rq}J_{1\mu}$$

$$J_{\mu+2} = \Phi_{1k} + N_{1((k-1)r+1)}J_{11} + N_{1(k-1)r+2}J_{12} + \cdots$$

$$+ N_{1q}J_{\mu+1}$$

$$\vdots$$

$$J_{r+1} = \Phi_{(r-\mu)k} + N_{(r-\mu)(k-1)r}J_{11} + N_{(r-\mu)(k-1)r+2}J_{12}$$

$$+ \cdots + N_{(r-\mu)q}J_{\mu+1}$$

$$J_{r+2} = \Phi_{(r-\mu+1)k} + N_{(r-\mu+1)(k-1)r+1}J_{11}$$

$$+ N_{(r-\mu+1)(k-1)r+2}J_{12} + \cdots + N_{(r-\mu+1)q}J_{\mu+1}$$

$$\vdots$$

$$J_{r+\mu+1} = \Phi_{rk} + N_{r((k-1)r+1)}J_{11} + N_{r((k-1)r+2}J_{12} + \cdots$$

$$+ N_{rq}J_{\mu+1}$$

$$J_{r+\mu+2} = \Phi_{1(k-1)} + N_{1((k-2)r+1)}J_{11}$$

$$+ N_{1((k-2)r+2}J_{12} + \cdots + N_{1q}J_{\mu+r+1}$$

$$\vdots$$

$$J_{2r+1} = \Phi_{(r-\mu)(k-1)} + N_{(r-\mu)((k-2)r+1)}J_{11}$$

$$+ N_{(r-\mu)((k-2)r+2}J_{12} + \cdots + N_{(r-\mu)q}J_{\mu+r+1}$$

$$J_{2r+2} = \Phi_{(r-\mu+1)(k-1)} + N_{(r-\mu+1)((k-2)r+1)}J_{11}$$

$$+ N_{(r-\mu+1)((k-2)r+2}J_{12} + \cdots + N_{(r-\mu+1)q}J_{\mu+r+1}$$

$$\vdots$$

$$J_{2r+\mu+1} = \Phi_{r(k-1)} + N_{r((k-2)r+1)}J_{11} + N_{r(k-2)r+2}J_{12}$$

$$+ \cdots + N_{rq}J_{\mu+r+1}$$

$$\vdots$$

$$J_{(k-1)r+\mu+2} = \Phi_{11} + N_{11}J_{11} + N_{12}J_{12} + \cdots$$

$$+ N_{1q}J_{(k-1)r+\mu+1}$$

$$\vdots$$

$$J_{kr+1} = \Phi_{(r-\mu)1} + N_{(r-\mu)1}J_{11} + N_{(r-\mu)2}J_{12}$$

$$+ \cdots + N_{(r-\mu)q}J_{(k-1)r+\mu+1}$$

$$J_{kr+2} = \Phi_{(r-\mu+1)1} + N_{(r-\mu+1)1}J_{11} + N_{(r-\mu+1)2}J_{12}$$

$$+ \cdots + N_{(r-\mu+1)q}J_{(k-1)r+\mu+1}$$

$$\vdots$$

$$J_{kr+\mu+1} = J_{q+1} = \Phi_{r1} + N_{r1}J_{11} + N_{r2}J_{12}$$

$$+ \cdots + N_{rq}J_{(k-1)r+\mu+1}.$$

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